Juniper Core Node BLR-CORE-01 QoS Congestion Event SOP (ALARM_2216)

Node: BLR-CORE-01 | **Vendor:** Juniper | **Severity:** Major | **Alarm Code:** ALARM_2216

Description

The Core Node BLR-CORE-01 is a Juniper device operating as a critical component of the network infrastructure. In this specific failure domain, the device is experiencing a QoS congestion event, characterized by microbursts, interface tail drops, and QoS queue starvation. This phenomenon can occur due to the inherent protocol mechanics of the Juniper's JUNOS operating system, which employs a sophisticated Quality of Service (QoS) framework to manage and prioritize traffic flows. The QoS framework operates across multiple layers of the protocol stack, including the Network Layer (Layer 3) and the Data Link Layer (Layer 2). When the device encounters an excessive number of packets or flows competing for bandwidth, the QoS system attempts to mitigate congestion by dynamically adjusting queue sizes, scheduling algorithms, and traffic policing mechanisms. However, under certain circumstances, these countermeasures can inadvertently exacerbate congestion, resulting in the observed symptoms. Furthermore, the Juniper's hardware components, such as the TCAM (Ternary Content-Addressable Memory) and the PFE (Packet Forwarding Engine), play a crucial role in processing and forwarding packets. In the event of hardware-related issues, the device may experience increased latency, packet loss, or dropped packets, contributing to the overall congestion event.

Pre-requisite Checks

- check: Verify that the Unknown Site is accessible and dispatch remote hands to investigate the issue

- check: Check the status of the redundant link to ensure it is operational and not contributing to the congestion event

- check: Review the network topology and configuration to identify any potential bottlenecks or misconfigurations

Potential Root Causes

Cause	Probability
QoS policy misconfiguration leading to queue starvation	High
Hardware-related issues affecting the TCAM or PFE	Medium
Excessive packet bursts or microbursts from upstream devices	Low

Diagnostic Commands & Expected Outputs

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Command: show system processes cpu, Expected Output: CPU usage and process information, Description: This command provides an overview of the device's CPU usage, allowing the technician to identify any potential issues related to processing power or resource allocation
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```
Command: show interfaces terse, Expected Output: Interface statistics and status, Description: This command displays detailed information about the device's interfaces, including statistics, status, and configuration details. This information can help identify any issues with interface utilization, errors, or misconfigurations

Command: show class-of-service policy summary, Expected Output: QoS policy summary and configuration, Description: This command provides a summary of the device's QoS policies, including configuration information and statistics. This information can help identify any issues with policy implementation, queue sizes, or scheduling algorithms
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Resolution Steps

Step 1: Step: Verify QoS policy configuration and adjust as necessary, Command: set class-of-service policy queue-depth , Description: This step involves reviewing the QoS policy configuration and adjusting the queue size or depth to ensure optimal performance and prevent queue starvation

Step 2: Step: Monitor and adjust hardware-related issues, Command: request system software info, Description: This step involves monitoring the device's hardware components, such as the TCAM and PFE, and adjusting the configuration or firmware as necessary to resolve any issues

Step 3: Step: Implement traffic shaping or policing mechanisms, Command: set class-of-service policy police , Description: This step involves implementing traffic shaping or policing mechanisms to regulate excessive packet bursts or microbursts from upstream devices

Post-Check Verification

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Command: show class-of-service policy summary, Expected Output: QoS policy summary and configuration, Description: This command provides a summary of the device's QoS policies, allowing the technician to verify that the configuration changes have been successfully implemented

Command: show interfaces terse, Expected Output: Interface statistics and status, Description: This command displays detailed information about the device's interfaces, allowing the technician to verify that the configuration changes have resolved any interface-related issues

Command: show system processes cpu, Expected Output: CPU usage and process information, Description: This command provides an overview of the device's CPU usage, allowing the technician to verify that the configuration changes have resolved any CPU-related issues
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: rollback class-of-service policy to version , Description: This step involves restoring the original QoS policy configuration to resolve any issues caused by the configuration changes

- Step: Revert hardware-related changes, Command: request system software info revert, Description: This step involves reverting any hardware-related changes made to resolve issues with the TCAM or PFE

- Step: Remove traffic shaping or policing mechanisms, Command: delete class-of-service policy police, Description: This step involves removing any traffic shaping or policing mechanisms implemented to regulate excessive packet bursts or microbursts from upstream devices